Joonsuk Bae

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Experimental high-energy nuclear physics — ALICE Run 3 jet physics & ePIC calorimetry R&D

Lead analyzer of the first ALICE Run 3 charged-particle jet measurement; Pb/scintillating-fiber calorimeter R&D for the ePIC Barrel Imaging Calorimeter.

Research Interests

- Inclusive and semi-inclusive jet cross sections, fragmentation, and substructure in pp and light-ion collisions with ALICE Run 3
- Jet quenching-like signatures and medium response in small systems (high-multiplicity pp, O–O, Ne–Ne)
- Non-perturbative QCD effects on jets: hadronization, multi-parton interactions, and CGC-inspired initial-state effects
- Imaging calorimetry for the EIC: beam-test characterization, simulation, and reconstruction of the ePIC Barrel Imaging Calorimeter (BIC)

Education

Ph.D. Candidate, Physics, Sungkyunkwan University, South Korea 2022.03 – expected 2026.08

Advisor: Beomkyu Kim
Thesis: *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE*

B.A., Physics, Sungkyunkwan University, South Korea 2021.02

Collaboration Membership

ALICE Collaboration, CERN, Geneva, Switzerland	2022.10 – present
ePIC Collaboration, Brookhaven National Laboratory, USA	2024.02 – present
LAMPS Collaboration, RAON, Korea	2022.03 – 2024.04

Research Experience

Lead analyzer, ALICE Jet Physics Working Group (PWG-JE), Run 3, CERN 2023.04 – present

- Leading the first ALICE Run 3 charged-particle jet cross-section analysis at $\sqrt{s} = 13.6$ TeV end-to-end: O^{2}Physics task development, QA, unfolding, systematics, and preliminary release; PWG-JE-approved results established the Run 3 inclusive charged-jet baseline.
- Designed the full Run 3 systematic-uncertainty program (tracking efficiency, track p_{T} resolution, unfolding stability, prior dependence) and established the Run 3-specific cross-section normalization based on the TVX visible cross section under continuous readout.
- Coordinated jet-enriched MC requests, train configurations, and data/MC quality assurance for the PWG-JE chain, including track–jet matching and detector-level closure tests.
- Presented the preliminary results as the ALICE official talk at **EPS-HEP 2025** and as posters at **Quark Matter 2025** and **Hard Probes 2024**; reported analyzer status regularly in PWG-JE working-group meetings (50+ internal talks since 2022).
- Extending the program to semi-inclusive hadron+jet measurements, R-dependent jet suppression, and jet substructure in small systems (high-multiplicity pp, O–O, Ne–Ne) as the basis of the thesis.

Detector physicist, ePIC Barrel Imaging Calorimeter (BIC), Korea-BIC consortium 2024.02 – present

- Led PMT-module QA prior to beam tests: signal characterization with cosmic-ray and radioactive-source data, timing-coincidence checks, optical-coupling stability, and per-module acceptance criteria.

- Led offline analysis of energy response, resolution, and linearity from two test-beam campaigns of the Pb/scintillating-fiber prototype (**KEK AF–AR** 2025.03, **CERN PS T10** 2025.07); analysis released as the first physics output of the Korea-BIC prototype.
- On-site responsibilities at both test beams included DAQ shifts, equalization, and prompt online analysis.
- Led JANA2 / EIC-recon reconstruction studies determining energy- and η-dependent sampling fractions for the BIC; contributed Geant4 prototype modeling.
- Participated in module fabrication and assembly: alignment of Pb/scintillating-fiber bundles, polishing of readout surfaces, and PMT/SiPM integration.

Detector R&D, LAMPS Forward Tracker, RAON, Korea 2022.03 – 2024.04

- Conducted early-stage design studies of an MPPC–scintillator-plane forward tracking detector, including sensor surveys, performance specifications, and cost trade-offs.
- Implemented preliminary Geant4 simulations of the Forward Tracker geometry as part of tracking-concept feasibility studies.

Selected Publications

Full publication list: inspirehep.net/authors/2117232
ALICE Collaboration co-author since October 2022 **(109 collaboration papers)**.

ALICE results led by the author (in preparation / internal review)

- FLAGSHIP ALICE Collaboration, *Inclusive and semi-inclusive charged-particle jet cross sections in pp collisions at* $\sqrt{s} = 13.6$ *TeV* (in preparation, target 2026); *first ALICE Run 3 charged-jet publication.*
  Role: lead analyzer; full responsibility for analysis, systematics, unfolding, and paper draft
- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at* $\sqrt{s} = 13.6$ *TeV with ALICE* — thesis analysis, foreseen 2026 publication.
  Role: lead analyzer; sole analyzer of the R-dependence component

Detector R&D and instrumentation (peer-reviewed)

- G. Cho, S. Kim, **J. Bae** *et al.*, “Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future e^+e^- colliders,” *J. Subatom. Part. Cosmol.* **3** (2025) 100021.
  Role: co-author; module construction and beam-test data taking
- H. Kim *et al.* (LAMPS), “Performance of the prototype beam drift chamber for LAMPS at RAON with proton and ^{12}C beams,” *JINST* **19** (2024) P12008, `arXiv:2412.08662`.
  Role: co-author; forward-tracking R&D and prototype performance studies
- Y. J. Kim, D. S. Ahn, J. K. Ahn *et al.* (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682.
  Role: co-author; Time-of-Flight detector maintenance and on-site checks
- LAMPS Collaboration, “Status of large acceptance multi-purpose spectrometer (LAMPS) at RAON,” *AIP Conf. Proc.* **2319** (2021) 080002.
  Role: co-author; SiPM/scintillator-based detector R&D

Publications in preparation (detector)

- Korea-BIC Collaboration, *Beam-test characterization of a Pb/scintillating-fiber imaging calorimeter prototype for the EIC* (3 papers in preparation, 2026).
  Role: primary data analyst across all three; co-lead author on the energy-response paper

Invited and Contributed Talks

Summary since 2022: **6 contributions to international conferences** (1 ALICE collaboration talk, 1 contributed talk, 4 posters) and **3 invited workshop talks**, with 5 invited collaboration-meeting talks and 7 domestic presentations.

Invited talks (workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, invited talk, *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* (2025.12)
- **1st Heavy-Ion Meeting (HIM 2025), APCTP**, invited talk, *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* (2025.05)
- **FKPPN Workshop**, invited talk, *Charged-jet studies in ALICE Run 3* (2023.11)

International conferences

- **EPS-HEP 2025, Marseille**, ALICE collaboration talk, *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (2025.07)
- **INPC 2025 (29th Int. Nuclear Physics Conf.), Daejeon**, contributed talk, *Charged-particle jet and jet quenching measurement in pp collisions at $\sqrt{s} = 13.6$ TeV during LHC Run 3 with ALICE* (2025.05)
- **Quark Matter 2025, Frankfurt**, poster, *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* (2025.04)
- **Hard Probes 2024, Nagasaki**, poster, *First measurements of charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (2024.09)
- **Quark Matter 2023, Houston**, poster, *Charged-particle multiplicity distribution in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE* (2023.09)
- **ATHIC 2023 (9th Asian Triangle Heavy-Ion Conf.), Incheon**, poster, *Dijet studies at the LHC* (2023.04)

Collaboration meetings (invited)

- **Korea-BIC 2025 Summer Workshop**, invited talk, *Data analysis of the recent beam test at CERN* (2025.09)
- **KoALICE National Workshop 2024**, invited talk, *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* (2025.01)
- **ALICE Physics Week 2024**, invited talk, *Charged-particle jet production in Run 3* (2024.12)
- **KoALICE National Workshop 2023**, invited talk, *Charged-jet O^2 MC and data QA* (2024.01)
- **KoALICE National Workshop 2022**, invited talk, *Dijet studies with ALICE at the LHC* (2023.01)

Domestic conferences (Korean Physical Society)

- **KPS Spring 2025** — contributed talk, *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at $\sqrt{s} = 13.6$ TeV with ALICE* (2025.04)
- **KPS Spring 2025** — poster, *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* (2025.04)
- **KPS Fall 2024** — contributed talk, *First jet measurement in ALICE Run 3: charged-particle jet production* (2024.10)
- **KPS Spring 2024** — contributed talk, *Charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV* (2024.04)
- **KPS Fall 2023** — contributed talk, *First look at charged-particle jet production in pp collisions at $\sqrt{s} = 13.6$ TeV with ALICE Run 3 data* (2023.10)
- **KPS Spring 2023** — poster, *Dijet studies at the LHC* (2023.04)
- **KPS Fall 2022** — poster, *Dijet studies with ALICE at the LHC* (2022.10)

Test-beam and Detector Activities

- **CERN PS T10** (2025.07) — ePIC BIC prototype: on-site analysis lead, energy-response and resolution studies.
- **KEK AF–AR** (2025.03) — ePIC BIC prototype: DAQ, equalization, prompt analysis.
- **Korea-BIC module production** (2024–2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA across all prototype modules.

Service and Professional Activities

- **ALICE PWG-JE jet physics working group** — regular analyzer presentations in weekly WG meetings; active contributor to O^2Physics jet task code base.
- **ePIC Korea-BIC analysis coordination** — primary analysis liaison between the SKKU group and the Korea-BIC consortium for beam-test data products.
- **KoALICE national activities** — regular student-level participation in KoALICE workshops since 2022.

Teaching and Mentoring

Student mentoring (non-supervisory), Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc. student) — ALICE Run 3 nuclear modification factor (R_{AA}) studies in small collision systems (O–O, Ne–Ne); guidance on O^2Physics framework and unfolding.
- *Harim Seong* (undergraduate) — detector-performance studies using JANA2-based reconstruction for the ePIC BIC; energy response and sampling-fraction extraction.

Teaching Assistant, Department of Physics, Sungkyunkwan University 2022 – 2024

- *Nuclear Reactions* (Graduate), Fall 2022
- *Quantum Mechanics II* (Undergraduate), Fall 2023
- *General Physics I* (Undergraduate), Spring 2022
- *General Physics Laboratory I* (Undergraduate), Spring 2022, 2023, 2024

Awards and Fellowships

- **BK21 FOUR Graduate Fellowship**, Ministry of Education / NRF Korea, SKKU Physics Education & Research Center — 2022–present.
- **Conference travel support**, Korea-CERN / KoALICE — Quark Matter 2023 (Houston), Hard Probes 2024 (Nagasaki), Quark Matter 2025 (Frankfurt), EPS-HEP 2025 (Marseille).

Technical Skills

Languages: English (fluent), Korean (native).
Programming: C++, Python, ROOT, LaTeX, Bash, Git.
HEP frameworks: ALICE O^2Physics, Geant4, JANA2 / EIC-recon, PYTHIA, FastJet.
Analysis methods: jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.
Detector hardware: PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

References

Available on request.